

LONDON GATWICK, United Kingdom

WMO: 037760

Lat: 51.148N

Lon: 0.190W

Elev: 62

StdP: 100.59

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	-2.6	-6.3	2.2	-1.5	-5.0	2.5	-1.0	11.7	9.4	10.4	8.9	1.3	70	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	27.7	19.1	25.4	18.0	23.8	17.4	19.9	25.6	18.9	23.9	18.0	22.2	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.0	22.0	17.1	12.3	20.6	16.2	11.6	19.6	56.9	25.7	53.7	24.0	51.0	22.0	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.4	DB	-7.0	30.7	1.6	2.2	-8.1	32.3	-9.1	33.5	-10.0	34.8	-11.2	36.3	(4)
(5)			WB	-7.2	21.7	1.6	1.1	-8.4	22.5	-9.3	23.1	-10.2	23.7	-11.4	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	4.9	5.2	7.0	9.4	12.4	15.5	17.7	17.4	14.6	11.5	7.8	5.4	(6)	
	DBStd	5.46	3.34	3.18	2.91	3.01	2.91	2.52	2.50	2.40	2.39	2.97	3.27	3.74	(7)	
	HDD10.0	697	159	137	99	46	9	0	0	0	1	19	81	147	(8)	
	HDD18.3	2815	416	369	352	268	183	91	42	48	114	213	317	401	(9)	
	CDD10.0	974	1	2	6	27	85	165	239	228	138	64	14	4	(10)	
	CDD18.3	50	0	0	0	0	1	6	23	18	2	0	0	0	(11)	
	CDH23.3	474	0	0	0	3	17	75	208	151	20	2	0	0	(12)	
	CDH26.7	96	0	0	0	0	1	12	46	36	1	0	0	0	(13)	
(14) Wind	WSAvg	3.4	3.9	3.9	3.8	3.5	3.5	3.3	3.1	3.0	3.0	3.2	3.2	3.6	(14)	
(15) Precipitation	PrecAvg	762	80	51	57	54	52	57	48	57	69	77	82	81	(15)	
	PrecMax	1058	183	143	118	113	142	137	103	131	224	196	196	146	(16)	
	PrecMin	508	14	7	4	4	1	7	10	1	12	3	15	15	(17)	
	PrecStd	104	44	33	27	30	29	35	26	32	50	49	45	37	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	14.1	17.2	23.0	25.2	28.5	30.1	30.1	25.8	21.0	16.0	13.9	(19)	
		MCWB	11.3	10.5	11.2	14.8	16.9	19.4	19.8	20.0	18.0	17.0	14.3	12.3	(20)	
	2%	DB	11.9	12.0	15.0	19.1	23.0	25.2	27.9	27.1	23.0	18.2	14.8	12.9	(21)	
		MCWB	10.5	9.9	10.7	12.9	16.3	17.5	19.1	19.5	17.2	15.2	13.4	11.7	(22)	
	5%	DB	11.0	11.0	13.1	16.9	20.8	23.1	25.8	24.8	20.9	17.0	13.9	12.0	(23)	
		MCWB	9.7	9.3	9.7	11.6	15.1	16.7	18.2	18.2	16.0	14.7	12.4	10.9	(24)	
	10%	DB	10.0	10.0	11.9	14.9	18.2	21.2	23.7	22.2	19.2	16.0	12.9	11.1	(25)	
		MCWB	9.1	8.7	9.1	10.6	13.5	15.8	17.4	16.9	15.3	14.1	11.4	9.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.5	12.4	15.3	18.6	20.4	21.2	21.5	19.5	17.3	14.9	12.7	(27)	
		MCDB	12.6	13.5	15.7	21.2	23.4	26.5	27.7	27.7	23.8	20.3	15.7	13.5	(28)	
	2%	WB	10.9	10.5	11.2	13.4	16.8	18.6	19.7	19.9	18.0	16.0	13.6	12.0	(29)	
		MCDB	11.7	11.6	13.9	18.2	21.3	23.8	25.9	25.3	21.5	17.7	14.6	12.8	(30)	
	5%	WB	10.0	9.6	10.5	12.2	15.4	17.4	18.8	18.7	17.1	15.3	12.6	11.2	(31)	
		MCDB	10.6	10.5	12.6	16.1	19.8	21.7	24.2	23.1	19.7	16.7	13.7	12.1	(32)	
	10%	WB	9.1	8.8	9.7	11.1	14.1	16.4	17.8	17.8	16.1	14.5	11.6	10.2	(33)	
		MCDB	9.8	9.7	11.3	14.1	17.8	20.3	22.3	21.4	18.5	15.9	12.8	11.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.3	7.6	8.9	9.2	9.4	9.4	9.3	8.9	7.6	6.3	5.5	(35)	
		MCDBR	5.6	7.0	10.4	12.4	13.3	13.3	13.6	13.0	11.5	8.1	6.0	4.7	(36)	
		MCWBR	5.4	5.4	6.8	7.1	7.4	6.9	6.2	6.1	6.4	5.6	5.2	4.6	(37)	
	5% WB	MCDBR	5.6	6.2	8.8	10.8	11.8	11.4	11.6	11.1	9.2	7.0	5.7	4.8	(38)	
		MCWBR	5.7	5.4	6.3	6.5	6.9	6.5	5.8	5.7	5.8	5.4	5.4	4.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.330	0.345	0.383	0.401	0.406	0.406	0.401	0.393	0.376	0.360	0.333	0.320	(40)	
	taud		2.371	2.376	2.267	2.238	2.271	2.296	2.315	2.357	2.397	2.422	2.416	2.361	(41)	
	Ebn at Noon		681	776	807	836	849	851	848	836	809	749	673	628	(42)	
	Edh at Noon		64	82	109	125	127	125	121	110	95	77	61	55	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.82	1.50	2.65	4.18	5.03	5.57	5.29	4.41	3.29	1.88	0.98	0.64	(44)	
	RadStd		0.07	0.20	0.34	0.42	0.32	0.51	0.46	0.34	0.26	0.17	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page